AeroTrust at a Glance

Avionics Telemetry Integrity Observatory & Verification System

Eval Tag: v0.3.2-verified

App Tag: v1.0.5-demo

SYNTHETIC DATASET NOTICE: All trajectories and threat injections are synthetic simulations. Populations: clean = neither spoofed nor jammed; non-spoofed = clean + jam-only. Scheduled legs N = 18,000 (pooled clean seeds 3001-3003): non-spoofed legs 12,986 (of which clean-only legs: 11,123), spoofed legs 5,014. No operational commercial flights are depicted.

1. EXECUTIVE SUMMARY

AeroTrust is an avionics data-integrity observatory designed to evaluate aircraft ADS-B telemetry against GNSS spoofing, interference, and atmospheric radiation anomalies in a single unified interface. Operating under frozen verification benchmarks on synthetic flight trajectories, it pairs physics-based altimetry consistency checks with an unsupervised abstention engine, classifying telemetry into four standardized evidence tiers (`no_alert` , `flagged_altitude_inconsistency` , `jamming_degraded` , `insufficient`). It features an aggregate Silent Acceptance Rate (SAR) auditor, geographic H3 interference mapping, and an Incident Scenario Calculator for aviation airworthiness directives.

2. OPERATIONAL CAPABILITIES & BOUNDARIES

✓ WHAT AEROTRUST CAN DO

- **Altimetry Verification:** Compares barometric vs GNSS altitude residuals, flagging vertical discrepancies > 120.0 m (Check a, Alert Rule; Checks b & c supporting).
- **Interference Clustering:** Aggregates multi-aircraft NIC/NACp/SIL drops into H3 spatial cells via dedicated Jamming Detector.
- **Evidence-Tiered Output:** Never exposes uncalibrated probabilities or claims flights are "safe"; uses standardized evidence tiers.
- **Privacy-Preserving SAR:** Evaluates Silent Acceptance Rate with cell suppression for cohorts where N < 30.
- **Incident Scenario Calculator:** Scenario arithmetic over public cases: aircraft-days held; money appears only if you enter a value.

✗ WHAT AEROTRUST CANNOT DO

- **0% Horizontal Spoofing Recall:** Cannot detect lateral track shifts or circular waypoint attacks (vertical only).
- **No Real-World Certification:** Benchmarks are synthetic; not calibrated on certified Flight Data Recorders.
- **Consensus Inactive:** Multi-aircraft consensus (Check d) is disabled due to self-referential graph loops.
- **Peacetime Base-Rate Drop:** At 0.01% threat prevalence, precision drops to 0.2% due to background barometric noise.
- **No Radiation Causality:** Solar proton flux and Kp indices serve strictly as illustrative exposure context, not a cause.
- **No Audited Financials:** Incident Scenario Calculator cases require verified primary AD documents (currently unchecked).

3. BENCHMARK RESULTS (SYNTHETIC DATA)

Claim	Reported value	Unit and seeds	Source table
Recall on all spoofed legs (pooled)	26.98% ([25.77%, 28.23%])	Scheduled flight legs, pooled seeds 3001–3003 (N = 5,014 spoofed legs / 18,000 legs; non-spoofed legs 12,986, of which clean-only legs: 11,123)	`docs/generated/base_rate_honesty_table.md & leg_level_metrics.md`
Recall by attack type	Altitude offset: 37.7%–51.6%; Combined: 34.0%–43.5%; Horizontal offset: 0.0%; Horizontal drift: 0.0% (Note: 40.8% is an earlier README figure that is no longer reported)	Scheduled flight legs per attack subtype, seeds 3001, 3002, 3003	`docs/generated/event_type_honesty_table.md`
Observed leg-level precision per seed	Seed 3001: 90.1% (TP=401, FP=44, denom=445); Seed 3002: 91.6% (TP=482, FP=44, denom=526); Seed 3003: 91.4% (TP=470, FP=44, denom=514) at benchmark attack share 26.6%–29.4% (1,595–1,763 spoofed legs / 6,001 legs per seed)	Scheduled flight legs per seed, unpooled (denominator: total flagged legs TP + FP)	`docs/generated/leg_level_metrics.md`
Expected precision at assumed threat prevalence (derived, assumed prevalence)	p=0.01%: 0.2% ([0.2%, 0.3%]); p=0.1%: 2.3% ([1.8%, 2.8%]); p=1%: 18.9% ([15.8%, 22.4%]); p=5%: 54.9% ([49.5%, 60.1%]); p=17.47%: 83.0% ([79.7%, 85.8%]) (Derived via Bayes rule from pooled TPR 26.98% and clean FPR 1.17%; 83.0% belongs here)	Derived via Bayes rule over scheduled flight legs (pooled seeds 3001–3003)	`docs/generated/base_rate_honesty_table.md`
Clean false positives per seed (v0.3.2 reproduced counts)	44, 44, 44 per seed (superseded '42 to 52' range removed). Breakdown: Seed 3001: 35 baro drift, 8 boundary truncation, 1 jamming; Seed 3002: 35 baro drift, 8 boundary truncation, 1 jamming; Seed 3003: 36 baro drift, 8 boundary truncation, 0 jamming	Clean scheduled flight legs per seed (TN: 4,193, 4,361, 4,300 across 6,001 legs per seed; clean FPR 1.17% denominator: 11,123 clean-only legs)	`docs/generated/fp_leg_audit_table.md`
Suppression and inactive/demo capabilities	SAR suppression: N >= 30 strictly enforced (20/24 regional cells masked); Horizontal recall: 0.0%; Consensus: Inactive (0% contribution, disabled in v0.3); Abstention: Benchmark: 0.0%. Rule-trigger demo: 125/200, which is by construction (demo scenarios were built to trigger rules), not a measured rate.	Regional/family cohorts and demo flights (seeds 3001–3003, demo seed 9999)	`docs/generated/sar_cell_suppression_table.md & abstention_demo_table.md`

At the benchmark's own leg-level attack share (about 27 to 29%), the derived and observed precision agree at about 90%. At assumed lower prevalences it falls as shown.

4. PLAIN-LANGUAGE DISCLAIMERS & KNOWN WEAKNESSES

1. **Altitude-consistency only:** Only vertical discrepancies trigger detection.
2. **False alarm base rate:** In peacetime airspaces, sensor turbulence produces ~117 false alarms per 10,000 clean flights.
3. **Abstention scope:** Benchmark commercial flights have N ≥ 68 points; abstention is exercised only in the demo dataset.
4. **Incident case presets:** All 3 presets (A320 Solar, A320 Panels, A330neo) display "primary source missing" in red until primary regulatory airworthiness directives are formally verified.